ZEDZAG B.V.

Tree'd History Guide

Museum Handset — Hardware Design Brief
& Requirements Specification

Prepared for: Electronics / Firmware Design Engineer
Product: Conversational AI audio guide handset (with microphone)
Confidential

Document	Handset Hardware Design Brief & Requirements Specification
Version	Draft v0.1 — for engineer review
Date	18 July 2026
Owner	Bahaa Hamadto, Founder / Director — ZedZag B.V.
Status	Working draft. Requirements marked Must / Should / Could.
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About this document

This document briefs an electronics/firmware engineer on the Tree'd History Guide handset so they can propose and design the device. It has two layers. The narrative sections (Parts 1–2) explain **what the product is and what the handset does** — read these first for context. The specification sections (Parts 3–7) give **numbered, testable requirements** to design against. Part 8 lists the open decisions we want the engineer's recommendation on, and Part 9 covers volumes, markets, and what we'd like back.

> **How to read the requirements**
> Every requirement has an ID (e.g. **PWR-03**) and a priority:
> **Must** — mandatory; the product does not work or ship without it.
> **Should** — strongly wanted; deviate only with a documented reason.
> **Could** — desirable if it fits cost, size, and schedule.
> *Targets are engineering goals, not final specs — challenge any that are impractical and propose alternatives.*

Contents

1. Product overview

1.1 What Tree'd is

Tree'd History Guide is an AI-powered conversational guide for museums and heritage sites. A visitor picks up a distinctive, museum-branded handset and converses with it freely — in any of 12+ languages — about the artwork or exhibit in front of them. There is no app, no screen, and no fixed script. The guide answers questions, follows the visitor's curiosity, and adapts to what they want to know. The result: visitors who once spent seconds with an artwork now spend minutes.

The AI is deployed locally at the museum and custom-trained on that museum's collection. It is designed to be energy-efficient and to run without any dependency on the public cloud during opening hours. Tree'd is already deployed in Cairo, Fayoum, and Amman, and is expanding across Europe (Netherlands, Belgium, UK) and MENA (Egypt, Jordan, GCC).

1.2 What the handset is

The handset is the visitor's physical point of contact with the guide — the one piece of Tree'd hardware a visitor ever touches. It is a custom, museum-branded device that a visitor holds like a telephone receiver: they speak into it and listen through it. It is handed out at the entrance, used throughout the visit, and returned and recharged at the end of the day. It must survive being handled by hundreds of visitors a day, every day, in a professional cultural venue.

> **Working ergonomic model (to confirm)**
>
> We assume a **telephone-handset form factor**: an earpiece (receiver) at one end held to the ear, and the microphone (mouthpiece) at the other end near the mouth — used like a phone, held to one ear.
>
> This is deliberate. Held-to-ear listening keeps audio **private** (visitors don't disturb each other), keeps the mic **close to the mouth** for a strong voice signal in noisy galleries, and physically separates the speaker from the mic to reduce echo. If you see a better ergonomic model, raise it — but the acoustic requirements below assume near-field, held-to-ear use.

1.3 What the handset does (and does not do)

Because the AI runs on an on-site server (see 1.4), the handset itself is a **smart audio endpoint**, not a computer that thinks. Its job is to capture the visitor's voice cleanly, stream it to the server, play back the response, and behave predictably with almost no controls.

The handset DOES	Capture voice via microphone · process/clean the audio locally (echo, noise, voice detection) · stream audio to the on-site server over Wi-Fi and receive the spoken reply · play clear audio to the ear · show simple status (LED / haptic) · run all day on battery · charge in a cradle · fail gracefully and tell the visitor when something's wrong
The handset does NOT	Run the AI / language model · store the collection · connect to the public internet for conversations · have a screen or touch UI · run an app · use a cellular/SIM connection · permanently store visitor audio

1.4 System architecture — where the AI runs

The conversational AI runs on an **on-site edge server** (located at the museum, on the local network). The handset does not perform AI inference. This keeps the handset light, low-power, low-cost, and long-lasting on battery, and concentrates the compute where it can be powered, cooled, and maintained. The engineer designs the handset to this model.

Conversation data path

1. Visitor speaks → handset microphone captures audio

2. Handset DSP → echo cancellation, noise suppression, voice-activity detection, encode/compress

3. Handset Wi-Fi → streams audio to the on-site edge server (museum LAN)

4. Edge server → speech recognition, the AI conversation, speech synthesis (all on-site)

5. Handset Wi-Fi → receives synthesized speech audio

6. Handset speaker → plays the reply to the visitor's ear

The network is on the **critical path of every conversation**. Wi-Fi coverage, latency, and reliability directly determine whether the product feels conversational. The handset must contribute as little latency as possible and behave sensibly when coverage drops (see Constraints, Part 6). The internet is used only for occasional model/firmware updates and logging — never for live conversation.

2. How the handset is used (functional description)

A typical day and interaction, described in plain language so the design serves the real workflow:

Start of day (staff)

- Staff take handsets from the charging cradle/rack where they charged overnight.
- Each handset powers on (or wakes) and automatically joins the museum Wi-Fi and connects to the on-site server. A status indicator shows it is ready.

A visitor conversation

- The visitor holds the handset to their ear like a phone.
- They begin to speak. **How the handset knows to listen is an open design decision** — a press-to-talk button is the robust baseline in a noisy gallery; hands-free voice detection is a possible enhancement (see Part 8).
- The handset indicates it is listening (e.g. LED colour and/or a short haptic pulse), captures the speech, and streams it to the server.
- Within a conversational delay, the spoken answer plays into the visitor's ear. The visitor asks a follow-up. This repeats naturally for the length of the visit.
- The visitor moves to the next exhibit and continues. The handset stays connected as they move through the galleries.

End of day (staff)

- Handsets are collected, wiped down/cleaned, and returned to the cradle to recharge for the next day.

> **Design implications of this workflow**
>
> Near-instant readiness on pick-up; automatic network + server connection with no visitor setup.
>
> Simplicity above all — most visitors are not technical and there is no screen to guide them. One obvious control at most.
>
> Robust roaming across Wi-Fi access points as visitors walk the galleries, without dropping the conversation.
>
> Fast, foolproof fleet charging and a surface that tolerates frequent cleaning.

3. Functional & technical requirements

Requirements are grouped by subsystem. Values are targets for the edge-server architecture; challenge any that are impractical.

3.1 Microphone & voice capture

ID	Requirement	Priority
MIC-01	Capture clear, intelligible speech from a visitor speaking into the mouthpiece at close range (near-field, held-to-ear use).	Must
MIC-02	Voice must be captured clearly with the mouthpiece at both **30 cm and 60 cm** from the mouth (per our QC standard), to tolerate varied handling.	Must
MIC-03	Use a digital MEMS microphone with high signal-to-noise ratio (target **SNR ≥ 64 dB(A)**) suited to speech-recognition front-ends.	Should
MIC-04	Capture wideband voice — sampling **≥ 16 kHz, ≥ 16-bit** — to give the server's speech recognition good input across 12+ languages.	Must
MIC-05	Optionally support a dual-mic array for beamforming / ambient-noise rejection if analysis shows a single mic is insufficient in loud galleries.	Could
MIC-06	Acoustic ports must resist dust and debris ingress and be cleanable without degrading capture.	Should

3.2 Speaker & audio output

ID	Requirement	Priority
SPK-01	Reproduce clear, natural speech to the ear via an earpiece/receiver, with no distortion at **60% and 80% volume** (per our QC standard).	Must
SPK-02	Support playback of synthesized speech across 12+ languages; reproduce the full speech band cleanly (playback path **up to 24 kHz**).	Must
SPK-03	Maximum output must be loud enough for hard-of-hearing visitors held to the ear, without harsh clipping. Volume must be adjustable by the visitor (see UI-04).	Must
SPK-04	Idle noise floor must be inaudible — no hiss, hum, or buzz when the handset is on but not playing (per QC).	Must
SPK-05	Earpiece placement and enclosure must minimise acoustic leakage so audio stays private to the user.	Should

3.3 Audio processing (codec / DSP)

ID	Requirement	Priority
DSP-01	Provide **acoustic echo cancellation (AEC)** so the microphone does not pick up the earpiece output — no feedback loop when speaker and mic are active together (per QC).	Must
DSP-02	Provide **noise suppression** tuned for gallery ambience (crowds, HVAC, reverberant halls) to improve the signal sent to the server.	Must

ID	Requirement	Priority
DSP-03	Provide **voice-activity detection (VAD)** so the handset streams only meaningful speech and can signal turn boundaries.	Should
DSP-04	Encode/compress captured audio for efficient, low-latency streaming while preserving speech-recognition quality.	Must
DSP-05	Confirm whether audio DSP is handled by a dedicated codec/DSP or integrated into the main SoC, and justify the choice on cost, power, and latency.	Should

3.4 Compute & firmware

ID	Requirement	Priority
CPU-01	Provide **only the compute needed for an audio endpoint** — audio pipeline, DSP, networking, control logic. AI inference is NOT on the handset.	Must
CPU-02	Recommend the platform (capable Wi-Fi MCU vs. small embedded-Linux SoC) with rationale on cost, power, latency, and update tooling.	Must
CPU-03	Support **secure over-the-air (OTA) firmware updates**, applied when the fleet is docked/idle, with rollback on failure.	Must
CPU-04	Report firmware version on boot and expose it for fleet management / QC (per QC).	Must
CPU-05	Boot to ready quickly on pick-up from the cradle; recover gracefully from crashes/errors without staff intervention.	Must

3.5 Connectivity

ID	Requirement	Priority
NET-01	Provide **Wi-Fi** (dual-band 2.4/5 GHz) as the link to the on-site server. No cellular/SIM.	Must
NET-02	Stream audio to the server and receive replies with **low, consistent latency**; buffer against network jitter without adding perceptible delay. Propose a handset latency budget.	Must
NET-03	Roam seamlessly between access points as the visitor moves through galleries, without dropping the conversation.	Must
NET-04	Support enterprise/museum Wi-Fi security (e.g. WPA2/WPA3) and a simple, secure per-venue provisioning method (no per-visitor setup).	Must
NET-05	Encrypt audio and control traffic **in transit** (visitor audio is personal data — see REG-05).	Must
NET-06	Optionally provide BLE for local provisioning, diagnostics, or cradle presence detection.	Could

3.6 Power & battery

ID	Requirement	Priority
PWR-01	Operate for a **full museum day on one charge** — QC minimum is 8 hours of use; **design target ≥ 10–12 hours** to cover long days and battery ageing.	Must
PWR-02	Recharge fully overnight (target **≤ 3–4 hours 0→100%**) so the whole fleet is ready each morning.	Must
PWR-03	Use a single rechargeable Li-ion/Li-Po cell with a protection circuit and safe charge management (temperature-aware).	Must
PWR-04	Recommend **cradle charging via exposed contacts (pogo pins)** for drop-in fleet charging, in preference to a plug-in port that wears and lets dust in. Include a serviceable port (e.g. USB-C) for bench/service use.	Should
PWR-05	No unexpected heat — no hot spots on the enclosure in normal use or on charge (per QC).	Must
PWR-06	Battery should be replaceable as a service part (design for a maintenance swap after cycle-life ageing), even if not user-accessible.	Should
PWR-07	Signal low battery clearly (indicator and/or audio prompt) before the handset dies mid-visit.	Must

3.7 Controls, indicators & feedback

ID	Requirement	Priority
UI-01	Keep the visitor interface **minimal** — there is no screen. At most one primary control (e.g. talk/power).	Must
UI-02	Provide a clear status indicator (e.g. **RGB LED**) for: ready, listening, thinking/streaming, low battery, charging, and error/offline.	Must
UI-03	Provide **non-visual confirmation of state** — e.g. a short haptic pulse and/or audio cue when the handset starts listening — since users hold it to the ear and cannot see the LED.	Should
UI-04	Provide a **volume control** the visitor can operate by feel (tactile, no screen).	Must
UI-05	Any controls must survive high-frequency daily use (hundreds of actuations/day) and be operable by visitors with limited dexterity.	Should

3.8 Enclosure & industrial design

ID	Requirement	Priority
ENC-01	Telephone-handset ergonomics (working assumption, 1.2): comfortable held to the ear, natural mic-to-mouth geometry, light enough for extended use.	Must
ENC-02	Custom, **museum-branded housing**. Colour and branding are specified per museum and supplied separately — design for in-mould/insert branding and colour variants; **do not hard-code a single colour**.	Must

ID	Requirement	Priority
ENC-03	Durable enough for daily public handling — survive repeated drops onto hard museum floors and heavy cycle use. Propose a drop-test target.	Must
ENC-04	Surfaces must be **wipe-clean and tolerate frequent disinfection** without degrading; consider antimicrobial material/coating.	Must
ENC-05	Dust/splash resistance appropriate to indoor galleries AND dusty open-air heritage sites (e.g. Fayoum). Propose an IP rating (e.g. IP54); reconcile with acoustic ports and charging contacts.	Should
ENC-06	Provide an attachment point for a lanyard/tether (loss prevention) that does not compromise durability or ergonomics.	Could
ENC-07	Materials should be manufacturable at our volumes (ABS / PC / PC-ABS) and RoHS/REACH compliant (see Part 5).	Must

3.9 Accessibility

Accessibility matters to museums and to the curators who champion us — build it in, not on.

ID	Requirement	Priority
ACC-01	Sufficient maximum volume and clear audio for hard-of-hearing visitors (see SPK-03), with easy tactile volume control (UI-04).	Must
ACC-02	Consider **hearing-aid compatibility** (e.g. telecoil / T-coil or equivalent) for the earpiece.	Could
ACC-03	Controls usable without sight and with limited dexterity; tactile landmarks to orient the device by feel.	Should
ACC-04	Weight and grip comfortable for children, older visitors, and extended holding.	Should

4. Operating & environmental conditions

The handset ships to both climate-controlled European galleries and hot, dusty MENA sites (Cairo, Fayoum, Amman, GCC). Design for the wider envelope.

Use environment	Indoor museum galleries and open-air / partially covered heritage sites
Operating temperature	Target 0 °C to 40 °C (accommodating open-air MENA sites); confirm with battery limits
Storage/transport temperature	Target −20 °C to 60 °C (vehicles/warehousing in hot climates)
Humidity	Operate across typical indoor and outdoor humidity; no condensation-driven failure
Dust	Tolerate dusty environments (see ENC-05); protect acoustic ports and charging contacts
Handling	Hundreds of pick-ups, drops, and cleaning cycles per unit per day
Duty cycle	Continuous availability across museum opening hours; overnight fleet charging

5. Compliance & standards

Because the handset contains a radio (Wi-Fi/BLE) and a lithium battery and sells into multiple regions, compliance affects the design from the start. Treat the list below as **design targets to confirm** with a compliance specialist at design freeze — regulations and standard versions change, and each market has its own radio type-approval process.

ID	Requirement	Priority
REG-01	EU (Netherlands, Belgium): **CE marking** including the Radio Equipment Directive (RED, 2014/53/EU) for the radio, EMC, and safety.	Must
REG-02	EU: **RoHS** (2011/65/EU), **WEEE**, and REACH for materials and e-waste.	Must
REG-03	UK: **UKCA** marking and equivalent UK radio/EMC/safety requirements.	Must
REG-04	Per-market **radio type approval** before deployment — e.g. Egypt (NTRA), Jordan (TRC), and GCC regulators (e.g. TDRA, CST). Plan lead time and cost per country.	Must
REG-05	Product safety for electronics (**EN/IEC 62368-1** class) and **battery safety (IEC 62133-2)**; UN 38.3 for lithium-battery transport.	Must
REG-06	Data protection: visitor audio is personal data under **GDPR**. Support encryption in transit (NET-05) and no persistent visitor-audio storage on the handset.	Must

6. Constraints & limitations

These are the boundaries and known trade-offs the design lives within. They are as important as the requirements — please design around them and flag any you think we should revisit.

6.1 Architectural constraints (edge-server model)

- **Server + network dependency:** the handset cannot converse on its own. If the on-site server or Wi-Fi is down, conversation stops. The handset must detect this and tell the visitor clearly (audio + indicator), not fail silently.
- **Wi-Fi coverage is mandatory everywhere the handset is used.** A gallery with a Wi-Fi dead spot is a gallery where the product fails. Coverage is partly the venue's responsibility, but the handset must roam well and degrade gracefully at the edges.
- **Latency is on the critical path.** Every conversation is a network round-trip to the server and back. The handset must add minimal latency and hide network jitter, or the experience stops feeling conversational.
- **No cellular fallback.** Connectivity is Wi-Fi only; there is no SIM/cellular backup if the venue network fails.

6.2 Physical & interaction limitations

- **No screen.** All feedback is audio, LED, and (ideally) haptic. Everything must be understandable without a display, by non-technical visitors, often held to the ear where the LED can't be seen.
- **Battery-bounded operating window.** Runtime caps continuous use; the fleet depends on reliable overnight charging and enough cradle capacity. A handset that doesn't last the day, or a cradle that charges too slowly, breaks the daily cycle.
- **Shared device hygiene.** Handsets pass between hundreds of people; surfaces must tolerate constant cleaning without wearing or clouding.
- **Acoustic reality.** Museums are noisy and reverberant. The held-to-ear form and DSP mitigate this, but no microphone fully removes gallery noise — capture quality (and therefore recognition accuracy) will always be worse in loud spaces.
- **Audio-bandwidth bound.** Multilingual playback quality is limited by the earpiece's acoustic performance.

6.3 Commercial & rollout constraints

- **Low initial volumes, scaling.** We start at tens of units per deployment and scale to hundreds. Design for cost-effective low-volume manufacture first, with a path to higher volumes — heavy tooling that only pays off at thousands of units is a risk early on.
- **Supply-chain resilience.** Prefer dual-sourced components and avoid single-source or end-of-life parts; long lead-time parts (>8 weeks) are a scheduling risk. LCSC/JLC-compatible parts are acceptable on cost-sensitive lines if dual-sourced.
- **Per-market approvals gate deployment.** Radio type approval (REG-04) must be scheduled and budgeted per country before we can ship there.
- **Pilot-driven timeline.** Our commercial motion is a 90-day pilot, so we need a small number of reliable, pilot-ready units relatively early — not a fully industrialised product on day one.

7. Acceptance criteria

A unit is accepted for a pilot/production batch when it passes Tree'd's handset QC. The design must make these outcomes reliably achievable. Full QC is a separate document; the gating checks are:

- Audio out: clear, undistorted playback at 60% and 80% volume; inaudible idle noise floor.
- Audio in: voice captured clearly at 30 cm and 60 cm; no feedback loop with speaker and mic active together.
- Power: charges correctly with correct indicator; extrapolated battery life exceeds 8 hours; no abnormal heat.
- Connectivity: joins the venue network and on-site server; roams without dropping; OTA/version check current.
- Firmware/behaviour: boots to the correct version; activates on the intended trigger; recovers gracefully from a forced error.
- Mechanical: enclosure and branding intact; controls and charging contacts/port sound; surfaces clean and unmarked.

> **Reliability bar**
> Target ≥ 90% first-pass QC yield per batch. A >10% batch failure rate stops the line and is escalated. Design for testability so units can be verified quickly and consistently in production.

8. Open decisions — we want your recommendation

These are deliberately unspecified. They are the core engineering trade-offs, and your recommendation (with reasoning) is exactly what we're looking for.

- **Turn-taking:** press-to-talk button vs. hands-free voice detection (or wake behaviour) in a noisy gallery — which is more robust and simpler for visitors?
- **Duplex:** half-duplex (listen, then speak) as baseline, or full-duplex with barge-in (visitor can interrupt the guide)? Impact on DSP/AEC and cost?
- **Compute platform:** capable Wi-Fi MCU vs. small embedded-Linux SoC — best balance of cost, power, latency, and update tooling for an audio endpoint?
- **Charging:** cradle contacts (pogo) vs. plug-in port as the primary method; cradle/rack design and units-per-cradle for a fleet.
- **Microphone:** single high-SNR mic vs. dual-mic array given near-field held-to-ear use — is the array worth the cost/complexity?
- **Codec/DSP:** dedicated audio codec/DSP vs. SoC-integrated audio — latency, power, and BOM implications.
- **Ingress protection:** target IP rating that reconciles indoor galleries with dusty open-air sites, acoustic ports, and charging contacts.
- **Battery:** capacity vs. weight trade-off to hit ≥10–12 h comfortably; serviceable-battery approach.

9. Production context & what we'd like back

9.1 Volumes, markets & timeline

Initial volumes	Tens of units per deployment, scaling to hundreds; fleet across multiple museums over time
Target markets	Netherlands, Belgium, UK (EU/UK compliance); Egypt, Jordan, GCC (MENA — heat/dust + per-country radio approval)
Commercial motion	90-day pilots — a small batch of reliable, pilot-ready units is needed relatively early
Sourcing preference	Dual-sourced, RoHS-compliant parts; avoid EOL/single-source and >8-week lead times
Cost	Per-unit cost target to be set with you — please provide preliminary BOM cost at 50 / 100 / 500 units

9.2 Requested deliverables

To take this forward, you’re expected to delivery a PCB design (ESP32 module, audio codec, charger, battery, LEDs, buttons).

> **Questions**
> Anything unclear or that you'd challenge — please raise it.
> Contact: **Bahaa Hamadto** · bahaa@treed.co